CUMULATIVE EXERCISE

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Two forms of elemental phosphorus: white and red.

The Many States of Phosphorus

Solid elemental phosphorus appears in a rich variety of forms, with crystals in all seven crystal systems reported under various conditions of temperature, pressure, and sample preparation.

(a) The thermodynamically stable form of phosphorus under room conditions is black phosphorus. Its unit cell is orthorhombic with edges of lengths 3.314, 4.376, and 10.48 Å. Calculate the volume of one unit cell, and determine the number of phosphorus atoms per unit cell, if the density of this form of phosphorus is 2.69 g cm^{-3}.

(b) The form of phosphorus that is easiest to prepare from the liquid or gaseous state is white phosphorus, which consists of P_4 molecules in a cubic lattice. When X-rays of wavelength 2.29 Å are scattered from the parallel faces of its unit cells, the first-order Bragg diffraction is observed at an angle 2θ of 7.10°. Calculate the length of the unit-cell edge for white phosphorus. At what angle will third-order Bragg diffraction be seen?

(c) Amorphous red phosphorus has been reported to convert to monoclinic, triclinic, tetragonal, and cubic red forms with different heat treatments. Identify the changes in the shape of the unit cell as a cubic lattice is converted first to tetragonal, then monoclinic, then triclinic.

(d) A monoclinic form of red phosphorus has been studied that has cell edge lengths 9.21, 9.15, and 22.60 Å, with an angle β of 106.1°. Each unit cell contains 84 atoms of phosphorus. Estimate the density of this form of phosphorus.

(e) Phosphorus forms many compounds with other elements. Describe the nature of the bonding in the solids white elemental phosphorus (P_4), black elemental phosphorus, sodium phosphate (Na_3PO_4), and phosphorus trichloride (PCl_3).

Answers

(a) Volume is 152.0 $Å^3$; eight atoms per unit cell

(b) 18.5 Å; angle $2\theta = 21.4°$

(c) Cubic to tetragonal: One cell edge is stretched or shrunk. Tetragonal to monoclinic: A second cell edge is stretched or shrunk, and the angles between two adjacent faces (and their opposite faces) are changed from 90°. Monoclinic to triclinic: The remaining two angles between faces are deformed from 90°.

(d) 2.36 g cm^{-3}

(e) P_4(white) and PCl_3 are molecular solids; P(black) is covalent; Na_3PO_4 is ionic.

PROBLEMS

Answers to problems whose numbers are boldface appear in Appendix G. Problems that are more challenging are indicated with asterisks.

Crystal Symmetry and the Unit Cell

1. Which of the following has 3-fold rotational symmetry? Explain.
(a) An isosceles triangle
(b) An equilateral triangle
(c) A tetrahedron
(d) A cube

2. Which of the following has 4-fold rotational symmetry? Explain.
(a) A cereal box (exclusive of the writing on the sides)
(b) A stop sign (not counting the writing)
(c) A tetrahedron
(d) A cube

3. Identify the symmetry elements of the CCl_2F_2 molecule (see Fig. 12.18).

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4. Identify the symmetry elements of the PF_5 molecule (see Fig. 3.23a).

5. The second-order Bragg diffraction of X-rays with λ = 1.660 Å from a set of parallel planes in copper occurs at an angle 2θ = 54.70°. Calculate the distance between the scattering planes in the crystal.

6. The second-order Bragg diffraction of X-rays with λ = 1.237 Å from a set of parallel planes in aluminum occurs at an angle 2θ = 35.58°. Calculate the distance between the scattering planes in the crystal.

7. The distance between members of a set of equally spaced planes of atoms in crystalline lead is 4.950 Å. If X-rays with λ = 1.936 Å are diffracted by this set of parallel planes, calculate the angle 2θ at which fourth-order Bragg diffraction will be observed.

8. The distance between members of a set of equally spaced planes of atoms in crystalline sodium is 4.28 Å. If X-rays with λ = 1.539 Å are diffracted by this set of parallel planes, calculate the angle 2θ at which second-order Bragg diffraction will be observed.

9. The members of a series of equally spaced parallel planes of ions in crystalline LiCl are separated by 2.570 Å. Calculate all the angles 2θ at which diffracted beams of various orders may be seen, if the X-ray wavelength used is 2.167 Å.

10. The members of a series of equally spaced parallel planes in crystalline vitamin B_{12} are separated by 16.02 Å. Calculate all the angles 2θ at which diffracted beams of various orders may be seen, if the X-ray wavelength used is 2.294 Å.

Crystal Structure

11. A crucial protein at the photosynthetic reaction center of the purple bacterium *Rhodopseudomonas viridis* (see Section 20.7) has been separated from the organism, crystallized, and studied by X-ray diffraction. This substance crystallizes with a primitive unit cell in the tetragonal system. The cell dimensions are $a = b$ = 223.5 Å and c = 113.6 Å.
    (a) Determine the volume, in cubic angstroms, of this cell.
    (b) One of the crystals in this experiment was box-shaped, with dimensions 1 × 1 × 3 mm. Compute the number of unit cells in this crystal.

12. Compute the volume (in cubic angstroms) of the unit cell of potassium hexacyanoferrate(III) ($K_3Fe(CN)_6$), a substance that crystallizes in the monoclinic system with a = 8.40 Å, b = 10.44 Å, and c = 7.04 Å and with β = 107.5°.

13. The compound $Pb_4In_3B_{17}S_{18}$ crystallizes in the monoclinic system with a unit cell having a = 21.021 Å, b = 4.014 Å, c = 18.898 Å, and the only non-90° angle equal to 97.07°. There are two molecules in every unit cell. Compute the density of this substance.

14. Strontium chloride hexahydrate ($SrCl_2{\cdot}6H_2O$) crystallizes in the trigonal system in a unit cell with a = 8.9649 Å and α = 100.576°. The unit cell contains three formula units. Compute the density of this substance.

15. At room temperature, the edge length of the cubic unit cell in elemental silicon is 5.431 Å, and the density of silicon at the same temperature is 2.328 g cm^{-3}. Each cubic unit cell contains eight silicon atoms. Using only these facts, perform the following operations.
    (a) Calculate the volume (in cubic centimeters) of one unit cell.
    (b) Calculate the mass (in grams) of silicon present in a unit cell.
    (c) Calculate the mass (in grams) of an atom of silicon.
    (d) The mass of an atom of silicon is 28.0855 u. Estimate Avogadro's number to four significant figures.

16. One form of crystalline iron has a bcc lattice with an iron atom at every lattice point. Its density at 25°C is 7.86 g cm^{-3}. The length of the edge of the cubic unit cell is 2.87 Å. Use these facts to estimate Avogadro's number.

17. Sodium sulfate (Na_2SO_4) crystallizes in the orthorhombic system in a unit cell with a = 5.863 Å, b = 12.304 Å, and c = 9.821 Å. The density of these crystals is 2.663 g cm^{-3}. Determine how many Na_2SO_4 formula units are present in the unit cell.

18. The density of turquoise, $CuAl_6(PO_4)_4(OH)_8(H_2O)_4$, is 2.927 g cm^{-3}. This gemstone crystallizes in the triclinic system with cell constants a = 7.424 Å, b = 7.629 Å, c = 9.910 Å, α = 68.61°, β = 69.71°, and γ = 65.08°. Calculate the volume of the unit cell, and determine how many copper atoms are present in each unit cell of turquoise.

19. An oxide of rhenium has a structure with a Re atom at each corner of the cubic unit cell and an O atom at the center of each edge of the cell. What is the chemical formula of this compound?

20. The mineral perovskite has a calcium atom at each corner of the unit cell, a titanium atom at the center of the unit cell, and an oxygen atom at the center of each face. What is the chemical formula of this compound?

21. Iron has a body-centered cubic structure with a density of 7.86 g cm^{-3}.
    (a) Calculate the nearest neighbor distance in crystalline iron.
    (b) What is the lattice parameter for the cubic unit cell of iron?
    (c) What is the atomic radius of iron?

22. The structure of aluminum is fcc and its density is ρ = 2.70 g cm^{-3}.
    (a) How many Al atoms belong to a unit cell?
    (b) Calculate a, the lattice parameter, and d, the nearest neighbor distance.

23. Sodium has the body-centered cubic structure, and its lattice parameter is 4.28 Å.
    (a) How many Na atoms does a unit cell contain?
    (b) What fraction of the volume of the unit cell is occupied by Na atoms, if they are represented by spheres in contact with one another?

24. Nickel has an fcc structure with a density of 8.90 g cm^{-3}.
    (a) Calculate the nearest neighbor distance in crystalline nickel.
    (b) What is the atomic radius of nickel?
    (c) What is the radius of the largest atom that could fit into the interstices of a nickel lattice, approximating the atoms as spheres?

25. Calculate the ratio of the maximum radius of an interstitial atom at the center of a simple cubic unit cell to the radius of the host atom.

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26. Calculate the ratio of the maximum radius of an interstitial atom at the center of each face of a bcc unit cell to the radius of the host atom.

Cohesion in Solids

27. Classify each of the following solids as molecular, ionic, metallic, or covalent.
(a) $BaCl_2$ (b) SiC
(c) CO (d) Co

28. Classify each of the following solids as molecular, ionic, metallic, or covalent.
(a) Rb (b) C_5H_{12}
(c) B (d) Na_2HPO_4

29. The melting point of cobalt is 1495°C, and that of barium chloride is 963°C. Rank the four substances in problem 27 from lowest to highest in melting point.

30. The boiling point of pentane (C_5H_{12}) is slightly less than the melting point of rubidium. Rank the four substances in problem 28 from lowest to highest in melting point.

31. Explain the relationship between the number of bonds that can be formed by a typical atom in a crystal and the possibility of forming linear, two-dimensional, and three-dimensional network structures.

32. Although large crystals of sugar (rock candy) and large crystals of salt (rock salt) have different geometric shapes, they look much the same to the untrained observer. What physical tests other than taste might be performed to distinguish between these two crystalline substances?

33. By examining Figure 21.17, determine the number of nearest neighbors, second nearest neighbors, and third nearest neighbors of a Cs^+ ion in crystalline CsCl. The nearest neighbors of the Cs^+ ion are Cl^- ions, and the second nearest neighbors are Cs^+ ions.

34. Repeat the determinations of the preceding problem for the NaCl crystal, referring to Figure 21.16.

Defects and Amorphous Solids

35. Will the presence of Frenkel defects change the measured density of a crystal?

36. What effect will the (unavoidable) presence of Schottky defects have on the determination of Avogadro's number via the method described in problems 15 and 16?

37. Iron(II) oxide is nonstoichiometric. A particular sample was found to contain 76.55% iron and 23.45% oxygen by mass.
(a) Calculate the empirical formula of the compound (four significant figures).
(b) What percentage of the iron in this sample is in the +3 oxidation state?

38. A sample of nickel oxide contains 78.23% Ni by mass.
(a) What is the empirical formula of the nickel oxide to four significant figures?
(b) What fraction of the nickel in this sample is in the +3 oxidation state?

A Deeper Look . . . Lattice Energies of Crystals

39. Calculate the energy needed to dissociate 1.00 mol of crystalline RbCl into its gaseous ions if the Madelung constant for its structure is 1.7476 and the radii of Rb^+ and Cl^- are 1.48 Å and 1.81 Å, respectively. Assume that the repulsive energy reduces the lattice energy by 10% from the pure Coulomb energy.

40. Repeat the calculation of problem 39 for CsCl, taking the Madelung constant from Table 21.5 and taking the radii of Cs^+ and Cl^- to be 1.67 Å and 1.81 Å.

41. (a) Use the Born–Haber cycle, with data from Appendices D and F, to calculate the lattice energy of LiF.
(b) Compare the result of part (a) with the Coulomb energy calculated by using an Li—F separation of 2.014 Å in the LiF crystal, which has the rock-salt structure.

42. Repeat the calculations of problem 41 for crystalline KBr, which has the rock-salt structure with a K—Br separation of 3.298 Å.

ADDITIONAL PROBLEMS

43. Some water waves with a wavelength of 3.0 m are diffracted by an array of evenly spaced posts in the water. If the rows of posts are separated by 5.0 m, calculate the angle 2θ at which the first-order "Bragg diffraction" of these water waves will be seen.

44. A crystal scatters X-rays of wavelength $\lambda = 1.54$ Å at an angle 2θ of 32.15°. Calculate the wavelength of the X-rays in another experiment if this same diffracted beam from the same crystal is observed at an angle 2θ of 34.46°.

45. The number of beams diffracted by a single crystal depends on the wavelength λ of the X-rays used and on the volume associated with one lattice point in the crystal—that is, on the volume V_p of a primitive unit cell. An approximate formula is

$$\text{number of diffracted beams} = \frac{4}{3}\pi\left(\frac{2}{\lambda}\right)^3 V_p$$

(a) Compute the volume of the conventional unit cell of crystalline sodium chloride. This cell is cubic and has an edge length of 5.6402 Å.
(b) The NaCl unit cell contains four lattice points. Compute the volume of a primitive unit cell for NaCl.
(c) Use the formula given in this problem to estimate the number of diffracted rays that will be observed if NaCl is irradiated with X-rays of wavelength 2.2896 Å.
(d) Use the formula to estimate the number of diffracted rays that will be observed if NaCl is irradiated with X-rays having the shorter wavelength 0.7093 Å.

46. If the wavelength λ of the X-rays is too large relative to the spacing of planes in the crystal, no Bragg diffraction will be seen because $\sin\theta$ would be larger than 1 in the Bragg equation, even for $n = 1$. Calculate the longest wavelength of X-rays that can give Bragg diffraction from a set of planes separated by 4.20 Å.

47. The crystal structure of diamond is fcc, and the atom coordinates in the unit cell are (0, 0, 0), $\left(\frac{1}{2}, \frac{1}{2}, 0\right)$, $\left(\frac{1}{2}, 0, \frac{1}{2}\right)$, $\left(0, \frac{1}{2}, v\right)$, $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$, $\left(\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right)$, $\left(\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right)$, and $\left(\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right)$. The lattice parameter is $a = 3.57$ Å. What is the C—C bond distance in diamond?

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48. Polonium is the only element known to crystallize in the simple cubic lattice.
    (a) What is the distance between nearest neighbor polonium atoms if the first-order diffraction of X-rays with $\lambda = 1.785$ Å from the parallel faces of its unit cells appears at an angle of $2\theta = 30.96°$ from these planes?
    (b) What is the density of polonium in this crystal (in g cm^{-3})?

49. At room temperature, monoclinic sulfur has the unit-cell dimensions $a = 11.04$ Å, $b = 10.98$ Å, $c = 10.92$ Å, and $\beta = 96.73°$. Each cell contains 48 atoms of sulfur.
    (a) Explain why it is not necessary to give the values of the angles α and γ in this cell.
    (b) Compute the density of monoclinic sulfur (in g cm^{-3}).

50. A compound contains three elements: sodium, oxygen, and chlorine. It crystallizes in a cubic lattice. The oxygen atoms are at the corners of the unit cells, the chlorine atoms are at the centers of the unit cells, and the sodium atoms are at the centers of the faces of the unit cells. What is the formula of the compound?

* 51. Show that the radius of the largest sphere that can be placed in a tetrahedral interstitial site in an fcc lattice is $0.225r_1$, where r_1 is the radius of the atoms making up the lattice. (*Hint:* Consider a cube with the centers of four spheres placed at alternate corners, and visualize the tetrahedral site at the center of the cube. What is the relationship between r_1 and the length of a diagonal of a face? The length of a body diagonal?)

52. What is the closest packing arrangement possible for a set of thin circular discs lying in a plane? What fraction of the area of the plane is occupied by the discs? Show how the same reasoning can be applied to the packing of infinitely long, straight cylindrical fibers.

53. Name two elements that form molecular crystals, two that form metallic crystals, and two that form covalent crystals. What generalizations can you make about the portions of the periodic table where each type is found?

54. The nearest-neighbor distance in crystalline LiCl (rock-salt structure) is 2.570 Å; the bond length in a gaseous LiCl molecule is significantly shorter, 2.027 Å. Explain.

55. (a) Using the data of Table 9.4, estimate the lattice energy and intermolecular separation of nitrogen in its solid state, assuming an fcc structure for the solid lattice.
    (b) The density of cubic nitrogen is 1.026 g cm^{-3}. Calculate the lattice parameter a and the nearest neighbor distance. Compare your answer with that from part (a).

* 56. Solid CuI_2 is unstable relative to CuI at room temperature, but $CuBr_2$, $CuCl_2$, and CuF_2 are all stable relative to the copper(I) halides. Explain by considering the steps in the Born–Haber cycle for these compounds.

57. A crystal of sodium chloride has a density of 2.165 g cm^{-3} in the absence of defects. Suppose a crystal of NaCl is grown in which 0.15% of the sodium ions and 0.15% of the chloride ions are missing. What is the density in this case?

58. The activation energy for the diffusion of sodium atoms in the crystalline state is 42.22 kJ mol^{-1}, and $D_0 = 0.145$ cm^2 s^{-1}.
    (a) Calculate the diffusion constant $D = D_0 \exp(-E_a/RT)$ of sodium in the solid at its melting point (97.8°C).
    (b) What is the root-mean-square displacement of an average sodium atom from an arbitrary origin after the lapse of 1.0 hour at $t = 97.8$°C? (*Hint:* Use Equation 9.28 in Chapter 9.)

59. A compound of titanium and oxygen contains 28.31% oxygen by mass.
    (a) If the compound's empirical formula is Ti_xO, calculate x to four significant figures.
    (b) The nonstoichiometric compounds Ti_xO can be described as having a Ti^{2+}—O^{2-} lattice in which certain Ti^{2+} ions are missing or are replaced by Ti^{3+} ions. Calculate the fraction of Ti^{2+} sites in the nonstoichiometric compound that are vacant and the fraction that are occupied by Ti^{3+} ions.

60. Classify the bonding in the following amorphous solids as molecular, ionic, metallic, or covalent.
    (a) Amorphous silicon, used in photocells to collect light energy from the sun
    (b) Polyvinyl chloride, a plastic of long-chain molecules composed of $-CH_2CHCl-$ repeating units, used in pipes and siding
    (c) Soda-lime-silica glass, used in windows
    (d) Copper-zirconium glass, an alloy of the two elements with approximate formula Cu_3Zr_2, used for its high strength and good conductivity

CUMULATIVE PROBLEMS

61. Sodium hydride (NaH) crystallizes in the rock-salt structure, with four formula units of NaH per cubic unit cell. A beam of monoenergetic neutrons, selected to have a velocity of 2.639×10^3 m s^{-1}, is scattered in second order through an angle of $2\theta = 36.26°$ by the parallel faces of the unit cells of a sodium hydride crystal.
    (a) Calculate the wavelength of the neutrons.
    (b) Calculate the edge length of the cubic unit cell.
    (c) Calculate the distance from the center of an Na^+ ion to the center of a neighboring H^- ion.
    (d) If the radius of an Na^+ ion is 0.98 Å, what is the radius of an H^- ion, assuming the two ions are in contact?

62. Chromium(III) oxide has a structure in which chromium ions occupy two-thirds of the octahedral interstitial sites in a hexagonal close-packed lattice of oxygen ions. What is the *d*-electron configuration on the chromium ion?

63. A useful rule of thumb is that in crystalline compounds every nonhydrogen atom occupies 18 $Å^3$, and the volume occupied by hydrogen atoms can be neglected. Using this rule, estimate the density of ice (in g cm^{-3}). Explain why the answer is so different from the observed density of ice.

64. Estimate, for the F-centers in CaF_2, the wavelength of maximum absorption in the visible region of the spectrum that will give rise to the color shown in Figure 21.29.

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